Bihar Engineering University, Patna

B.Tech 5th Semester Examination, 2024

Course: B.Tech
Code: 101501

Subject: Analysis and Design of Concrete Structure

Time: 03 Hours
Full Marks: 70

Instructions:-

(i) The marks are indicated in the right-hand margin
(ii) There are NINE questions in this paper
(iii) Attempt FIVE questions in all
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct option / answer the following (Any seven question only): [2 x 7 = 14]

(a) The stress-strain curve of mild steel shows a distinct yield point in
- (i) Cold-formed steel
- (ii) High strength deformed bars
- (iii) Hot-rolled steel
- (iv) Cast iron

(b) In limit state method, the partial safety factor for concrete is
- (i) 1.0
- (ii) 1.5
- (iii) 1.15
- (iv) 2.0

(c) In a singly reinforced rectangular beam, reinforcement is provided
- (i) Only in compression zone
- (ii) In both tension and compression zones
- (iii) Only in tension zone
- (iv) None of the above

(d) Two-way slabs are preferred over one-way slabs when the
- (i) Longer span is more than twice the shorter span
- (ii) Both spans are equal
- (iii) Shorter span is more than twice the longer span
- (iv) Ratio of longer span to shorter span is less than 2

(e) The Young's Modulus of steel is approximately
- (i) 2×10^5 MPa
- (ii) 2×10^3 MPa
- (iii) 2×10^4 MPa
- (iv) 2×10^6 MPa

(f) Development length increases with
- (i) Increase in bar diameter
- (ii) Decrease in bond strength
- (iii) Increase in stress
- (iv) All of the above

(g) The minimum percentage of steel in slab as per IS 456 is
- (i) 0.10%
- (ii) 0.15%
- (iii) 0.12%
- (iv) 0.20%

(h) In columns, lateral ties are provided to
- (i) Resist bending
- (ii) Confine concrete and hold longitudinal bars
- (iii) Improve aesthetics
- (iv) Improve finish

(i) Flanged beams are commonly found in
- (i) Slabs
- (ii) T-beams
- (iii) Columns
- (iv) Foundations

(j) The effective span of a simply supported beam is
- (i) Clear span + effective depth
- (ii) Clear span
- (iii) Clear span + width of support
- (iv) Lesser of clear span + effective depth or center-to-center of support

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Q.2 (a) Derive the moment of resistance for a singly reinforced rectangular beam. [7]

(b) Design a simply supported rectangular beam having Clear span = 4 m, Dead load = 10 kN/m, Live load = 15 kN/m. Use M20 concrete and Fe 415 steel. Assume Width = 230 mm. Provide depth and reinforcement with neat sketch. [7]

Q.3 (a) What are the limit states considered in structural design? Explain each with examples. [7]

(b) Discuss the failure modes in shear and how design for shear is performed. [7]

Q.4 (a) Define bond and explain how development length is calculated. [7]

(b) A simply supported reinforced concrete beam of span 5 m is subjected to a factored uniformly distributed load of 30 kN/m (including self-weight). The beam has an effective depth of 500 mm, and the reinforcement provided is 4 bars of 16 mm diameter of Fe 415 steel. The concrete grade used is M25. The beam is simply supported, and the tension reinforcement is terminated at the support. Calculate the development length. [7]

Q.5 (a) Differentiate between one-way and two-way slab with sketches. [7]

(b) Design a one-way slab having Room size = 3 m × 6 m, Live load = 3 kN/m², Floor finish = 1 kN/m², Use M20 and Fe 415 steel. Provide thickness, spacing and bar diameter with neat sketch. [7]

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Q.6 (a) Differentiate between short column and long column. How does it affect design? [7]

(b) Design a square column having Axial load = 800 kN, Column height = 3 m. Use M25 concrete and Fe 415. Provide size, number of bars, lateral ties with neat sketch. [7]

Q.7 (a) Write a step-by-step procedure to design a combined footing. [7]

(b) Design an isolated footing for a column having Load = 900 kN, Safe bearing capacity = 200 kN/m². Use M25 and Fe 415. Provide plan size, depth, and reinforcement with neat sketch. [7]

Q.8 (a) Explain the design of dog-legged staircase with diagram. [7]

(b) Design a dog-legged reinforced concrete staircase for a residential building with the following data: Clear floor-to-floor height = 3.3 m, Staircase is enclosed between walls with support at ends. Width of stair = 1.2 m, Tread = 250 mm, Riser = 165 mm, Width of landing = 1.2 m (equal to stair width), Live load on stair = 3 kN/m², Floor finish = 0.75 kN/m². Use M25 concrete and Fe 415 steel. Flight spans longitudinally (between landings) = 3 m (effective). Design the waist slab for the staircase and show the steps clearly. [7]

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Q.9 Write short notes on ***any two*** of the following: [2 x 7 = 14]

(a) Types of loads considered in building design.

(b) Stress-Strain Relationship of Concrete.

(c) Stress block parameters in RCC

(d) Safety in design

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